



Full length article

Uncovering neutral and adaptive genomic differentiation among European perch with brackish water and freshwater origin in the western Baltic Sea region

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ABSTRACT

Environmental variation across the range of wild species can lead to local adaptations. The Baltic Sea was formed when the Fennoscandian ice sheet retreated around 12 thousand years ago, creating a new brackish water habitat colonised by both marine and freshwater fish species. The European perch (*Perca fluviatilis*) is a predatory freshwater fish with a large geographical distribution across Eurasia, where it inhabits a wide range of environmental niches. In the Baltic Sea region it has even developed a specialised brackish water perch variant that can tolerate environmental salinity levels, which are lethal to freshwater perch. However, very little is known about the underlying mechanisms facilitating the colonisation and adaptation of perch to the Baltic Sea. Here, we use Genotyping-By-Sequencing data from six freshwater and six brackish water localities to disclose the evolutionary relationship between freshwater and brackish water perch. Our results show that the brackish water perch occurs in multiple distinct genetic clusters. We find that gene flow between brackish water perch with full access to the sea likely led to lower levels of differentiation and higher diversity than in freshwater perch. Selection analyses suggest that genomic adaptation played a role in the colonisation of the Baltic Sea and that the top three regions under selection harbour salinity tolerance genes. We complete by discussing the implications of our findings for management of brackish water perch in the western Baltic sea.

1. Introduction

Environmental heterogeneity plays a key role in shaping intraspecific variation in wild species (McDonald and Ayala, 1974). Novel habitats can offer new opportunities and challenges to populations in a subsection of the species distribution range, thus promoting local adaptation and increased levels of differentiation (Langerhans et al., 2013). Hence, intraspecific variation in morphology, behaviour and physiological capabilities is often correlated with environmental variation and can over time lead to the formation of separate ecotypes (Foote

et al., 2016; Skovrind et al., 2016). While phenotypic plasticity can be a vector towards adaptation (Radersma et al., 2020), genomic diversification is often the key mechanism governing the formation of ecotypes (Feder et al., 2012; Foote et al., 2016; Seehausen et al., 2014). When populations inhabit separate niches and no longer interbreed, genetic drift and/or selection will affect them separately and increase the differentiation between them across the genome and in particular in genes underpinning local adaptation.

Almost all teleost species have internal osmotic pressures corresponding to salinities of 9–12 parts per thousand (ppt) (Brett, 1979), but

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